

BIOSTAT 718

Homework Assignment 02

Due: Sunday evening, February 2, 2025

Crouse (Am J Cardiol, 2001) noted that intima-media thickness (IMT) of the carotid arteries is a generalized index of atherosclerotic disease. Previous investigators (Spence JD, Am J Cardiol 2002; Folsom AR, Stroke 1993; Howard G, Stroke 1993) have reported that IMT ranges between 0.5 to 1 mm in the general population and increases approximately 0.010 to 0.018 mm/year. Higher values of IMT are associated with more diseased arteries and cardiovascular events. These investigators and others have noted that lifestyle changes or medications may slow, arrest or induce regression in IMT.

An investigation of an experimental statin – a medication for individuals at high risk of having cardiovascular disease events (e.g., myocardial infarction, stroke) – was conducted to evaluate the effect of the drug on intima-media thickness (IMT) of the carotid arteries over a one-year period. In the prospective interventional study, approximately 2000 older patients (ages between 65-95 years) were randomized equally in a blinded fashion to receive either 40 mg of an experimental statin or a placebo. Treatments were administered daily for a year.

The intima-media thickness (IMT) of the common carotid arteries were measured by ultrasonography and were collected at two times for the study: prior to the drug or placebo administration (Year 0), and one year later (Year 1).

The data are provided in an Excel file [imtdata.xlsx](#) on the class website. The study variables are:

id: subject id number
trt: statin treatment (0=placebo, 1=40 mg experimental statin)
age: age of subject in years
imt0: IMT measurement at Year 0 (baseline)
imt1: IMT measurement at Year 1 (end of study)
imtdif: difference in one-year IMT measurements
male: male sex (0=female, 1=male)
smk: current smoking status (0=No, 1=Yes)

The dataset is formatted with one record per patient. There are no missing values.

Question 1: Please investigate whether there is a treatment effect for the experimental statin by considering three models for the study endpoint (outcome):

POST: Year 1 IMT outcome (IMT1)

CHANGE: Difference in Year 1 and Year 0 outcomes (IMT1 – IMT0)

ANCOVA: Year 1 IMT outcome, adjusting for Year 0 IMT outcome

- (a) Fit separate models for POST, CHANGE, and ANCOVA versus the treatment indicator variable (trt: 0=placebo, 1=40 mg statin). Summarize the key estimates in a single table. Report the results in three short, separate paragraphs (one paragraph for each model). Please be sure to interpret your estimates, interval estimates, and provide p-values and your decisions.
- (b) Do any of the models provide valid estimates of a treatment effect? Please explain why or why not.
- (c) For a pre-planned analysis to investigate the scientific question under the described study design, please state which model you would have used to investigate the question and justify your chosen model. (You can only choose one of the three.)

Question 2: One of the study investigators asked you, the study biostatistician, to perform analyses like the ones in Question 1. You are asked to determine whether IMT differs by sex. You are asked to fit the POST, CHANGE, and ANCOVA models noted above, but substituting in the male sex indicator variable (male: 0=female, 1=male) in place of the treatment indicator variable (trt: 0=placebo, 1=40 mg statin).

- (a) Fit separate models for POST, CHANGE, and ANCOVA as prescribed. Summarize the key estimates in a single table.
- (b) Interpret the estimated regression coefficient for the male sex variable for each of the three models fitted in part (a). Are the estimates approximately the same in magnitude? Are the interpretations the same for the three models? Please explain why or why not.
- (c) If the study investigator who asked you to perform the analyses mentioned their objective was to learn if IMT differs over time by sex, would the three models be acceptable to address the investigator's objective? If not, which model or models would be most appropriate? Briefly explain.

This assignment is due by midnight, February 2, 2025. Please upload your assignment to the class website. Please save your work in the MS Word format (so that it can be edited).